
A CONTEXT-AWARE SHARED-MEMORY FABRIC FOR COLLABORATIVE MOBILE ROBOTS

ABSTRACT

We propose a freshness-bounded distributed shared memory (DSM) for warehouse-scale multi-agent systems that exposes tile-level reads and writes and supports atomic task claims. The DSM combines Conflict-Free Replicated Data Type (CRDT) based merges with an owner-halo gossip protocol to bound the age of information, reduce task propagation time, and lower tail completion time compared to a centralized memory server at equal bandwidth, independent of per-agent control policies. Our design is guided by stigmergic communication and is locality driven, time constrained, and merge safe, with explicit token budgets to enforce bandwidth discipline. We position swarm intelligence as the collective context layer and deep neural networks as per-agent cognition, and show that a freshness-safe, conflict-free, bandwidth-bounded memory enables the swarm to behave coherently as one system. Ablations compare centralized versus DSM owner coordination systems on propagation, completion time and stability. We assess halo gossip’s impact on border oscillations and cross-region propagation, and evaluate merge operators for stability under high-density workloads. This work reframes the environment’s global memory as a systems advance for embodied multi-agent coordination, turning simple, local behaviors into scalable, robust collective performance.

1 INTRODUCTION

Warehouses with mobile robots require scalable coordination to assign and dispatch tasks. We study the effect of coordination architecture on task propagation delay, time to completion, and scalability. We present a compact analytical model that separates the physical travel plane from the memory/coordination plane.

2 BACKGROUND

A. Coordination Systems

Recent advances in multi-agent systems have demonstrated the potential application in fleets of mobile robots to automate warehouse operations, including item picking, sorting, and transportation (Agaskar et al., 2025). Effective coordination among robots is crucial to ensure efficient task allocation, collision avoidance, and overall system performance.

We build on queueing theory for edge and node delays, specifically using Kingman’s approximation for G/G/1 queues, age-of-information (AoI) bounds for gossip-based distributed shared memory, and standard centralized batch allocation models.

B. Stigmergy and Distributed Coordination

Stigmergy, a mechanism of indirect coordination through the environment, has been widely studied in swarm robotics (??). In warehouse settings, stigmergic ap-

proaches can enable robots to share information about task status and environmental changes without centralized control (??). This can lead to more robust and scalable systems, as robots can adapt to dynamic conditions and locally optimize their behavior. A successful implementation requires a well-designed shared memory and communication architecture.

C. Inference Systems for Embodied Robotic Agents

Besides coordination, mobile robots often rely on on-board AI models for perception, navigation, and decision-making. Recent work has explored the deployment of deep learning models on resource-constrained devices, enabling real-time inference for tasks such as object detection, semantic segmentation, and path planning (??). Vision language Action Models (VLMs) have shown promise in enabling robots to understand and execute complex instructions by grounding language in visual perception (??). However, deploying these models on mobile robots presents challenges related to computational efficiency, latency, and energy consumption. Device-aware edge inference SDKs, such as Nexa, provide efficient and portable execution of AI models on heterogeneous hardware. However, these SDKs do not address multi-agent task dissemination, claiming, or AoI-bounded coordination. Our contribution is a network-centric orchestration approach: a trace and signal dissemination layer combined with a local claiming policy that reduces task propagation time and increases the stability region compared to centralized and hybrid baselines, independent of the specific per-device

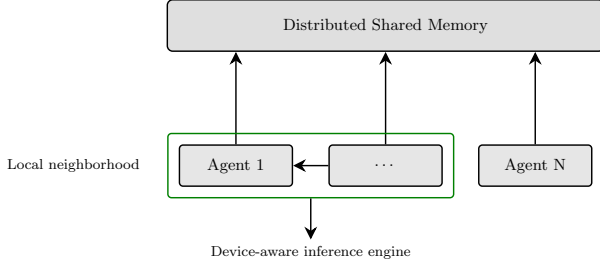


Figure 1: High-level architecture: agents coordinate via a distributed shared memory fabric while running local device-aware inference.

model runtime.

We validate this approach across multiple runners (TensorRT, ORT, Nexa) to demonstrate that the results are determined by the coordination architecture rather than the SDK implementation.

3 METHODOLOGY

We present an analytical and simulation models for coordination and distributed shared memory (DSM) architectures in warehouse graphs. The models compares centralized batch allocation with a DSM memory-agent approach, and derives propagation, claim, and queueing delay components to predict end-to-end order response times and stability conditions.

4 ANALYTICAL MODEL

Physical plane. The warehouse is modeled as a directed graph $G = (V, E)$. Each edge $e \in E$ has length L_e and *free-flow traverse time* $t_{\text{traverse},e} = L_e/v$ at speed v . Edges and intersections have *throughput capacity* μ_e (robots/s) and incur *latency due to traffic* L_e^{wait} when demand approaches capacity (utilization $\rho_e = \lambda_e/\mu_e$). A robot's travel time sums per-edge wait and traverse times. The same applies to work nodes (pickups, dropoffs) with service rate μ_n and wait L_n^{wait} .

Compute plane. The compute plane manages coordination and memory for task assignment. We consider two approaches: centralized memory-server, and distributed shared memory (DSM) with regional tile owners. In DSM, agents maintain shared state over warehouse tiles, ensuring data freshness via age-of-information (AoI) constraints.

Notation. N : number of robots; λ : order arrival rate (orders/s); S : per-order service time excluding propagation/claim; T_{prop} : order propagation delay; T_{claim} : claim/commit delay; T_{travel} , T_{node} : travel and node work times; $t_{\text{traverse},e}$, L_e^{wait} : traverse and wait time for edge e ;

$t_{\text{work},n}$, L_n^{wait} : work and wait time at node n ; $c_{\text{arr},e}$, $c_{\text{trav},e}$: coefficient of variation; AoI: age-of-information; τ : AoI budget; ρ_e , ρ_n : utilization.

Baseline service time

Orders arrive as a Poisson process with rate λ . For an order assigned to robot r , the baseline service time is

$$S = T_{\text{travel}} + T_{\text{node}}. \quad (1)$$

Propagation models

Centralized control system: Propagation time is modeled with accounting for batching, solver and broadcast time. Let Δ_B be the server batch period, $T_{\text{solve}}(N)$ the allocator runtime, and D_{tree} the height of the broadcast tree with per-hop delay δ_{hop} :

$$T_{\text{prop}}^{\text{cent}} \approx \Delta_B + T_{\text{solve}}(N) + D_{\text{tree}} \delta_{\text{hop}}. \quad (2)$$

Distributed Control System: we model gossip among memory agents and local tiles. Let H_{own} be owner-owner hop distance from the order's seed to the nearest eligible region, and f the gossip fanout (rumor spreading among owners/agents):

$$T_{\text{prop}}^{\text{dsm}} \approx (H_{\text{own}} + \log_f N) \delta_{\text{hop}}. \quad (3)$$

(When owner placement follows graph-Voronoi, H_{own} scales with graph diameter rather than N .)

Claim delay

To account for local contention, we use continuous random backoff, each of the k contenders draws $B_i \sim \text{Uniform}(0, W)$; the earliest one wins:

$$\mathbb{E}[T_{\text{claim}}] = \mathbb{E} \left[\min_i B_i \right] + \text{RTT} = \frac{W}{k+1} + \text{RTT}. \quad (4)$$

(Under centralized assignment T_{claim} is typically negligible vs. T_{solve} .)

Queueing Models

We use Kingman's G/G/1 queueing to model edge and node wait times as follows. For a path $\mathcal{P}$ through edges and nodes:

$$T_{\text{travel}} = \sum_{e \in \mathcal{P}} (t_{\text{traverse},e} + L_e^{\text{wait}}), \quad (5)$$

$$T_{\text{node}} = \sum_{n \in \{\text{pickup, drop, ...}\}} (t_{\text{work},n} + L_n^{\text{wait}}). \quad (6)$$

Edge wait time:

$$L_e^{\text{wait}} \approx \frac{\rho_e}{1 - \rho_e} \cdot \frac{c_{\text{arr},e}^2 + c_{\text{trav},e}^2}{2} \cdot \frac{1}{\mu_e},$$

where $\mu_e \approx \frac{1}{t_{\text{traverse},e}}$, $\rho_e = \frac{\lambda_e}{\mu_e} \in (0, 1)$. (7)

Node wait time:

$$L_n^{\text{wait}} \approx \frac{\rho_n}{1 - \rho_n} \cdot \frac{c_{\text{arr},n}^2 + c_{\text{work},n}^2}{2} \cdot \frac{1}{\mu_n},$$

where $\mu_n \approx \frac{1}{t_{\text{work},n}}$ (or $\frac{c}{t_{\text{work},n}}$ for c bays),

$$\rho_n = \frac{\lambda_n}{\mu_n} \in (0, 1). \quad (8)$$

Age of Information (AoI) constraint

Decisions consume tiles only if $\text{AoI} \leq \tau$. For periodic owner gossip with period T_g , the mean AoI is $\bar{A} \approx T_g/2$; choose $T_g \leq 2\tau$ (enforced token buckets) so AoI violations remain small.

Total response time and tails

Per order:

$$T_{\text{total}} = T_{\text{prop}} + T_{\text{claim}} + T_{\text{travel}} + T_{\text{node}}. \quad (9)$$

We evaluate performance with T_{50} and T_{90} from empirical and fitted distributions with a lognormal fit to T_{total} .

Stability

Let λ be the total order arrival rate (orders/s), N the number of robots, and μ_{eff} the effective per-robot service rate (orders/s):

$$\mu_{\text{eff}} = \frac{1}{\mathbb{E}[T_{\text{travel}} + T_{\text{node}}]}, \quad (10)$$

where T_{travel} sums edge wait + traverse, and T_{node} sums node wait + work.

$$\lambda < N \cdot \mu_{\text{eff}}. \quad (11)$$

Thus: arrivals must be less than fleet capacity.

Every busy edge/node must satisfy:

$$\rho_e = \frac{\lambda_e}{\mu_e} < 1, \quad \rho_n = \frac{\lambda_n}{\mu_n} < 1, \quad (12)$$

where $\mu_e \approx 1/t_{\text{traverse},e}$ (or $c/t_{\text{traverse},e}$ for c lanes) and $\mu_n \approx 1/t_{\text{work},n}$ (or $c/t_{\text{work},n}$ for c bays). If any $\rho \rightarrow 1$, waits blow up and μ_{eff} drops.

Architectures with smaller T_{prop} and fewer AoI violations typically start service sooner and increase μ_{eff} .

Comparative predictions

- For small N , (2) may be competitive; as N grows, $T_{\text{solve}}(N)$ and D_{tree} dominate, while (3) grows with $\log_f N$ and graph diameter $\Rightarrow$ DSM reduces T_{prop} .
- Lower T_{prop} reduces queue arrival bursts, lowering ρ_e on critical edges and hence L_e^{wait} , improving T_{90} .

Table 1: System parameters for model evaluation.

Parameter	Value	Units
δ_{hop}	[5, 20]	ms
Δ_B	[100, 300]	ms
f	{1, 2, 4}	hops
v	1.0	m/s
μ_e	$1/t_e^{\text{pass}}$	1/ms
τ_{tile}	300	ms

Analytical Results

We summarize three key results from the analytical study.

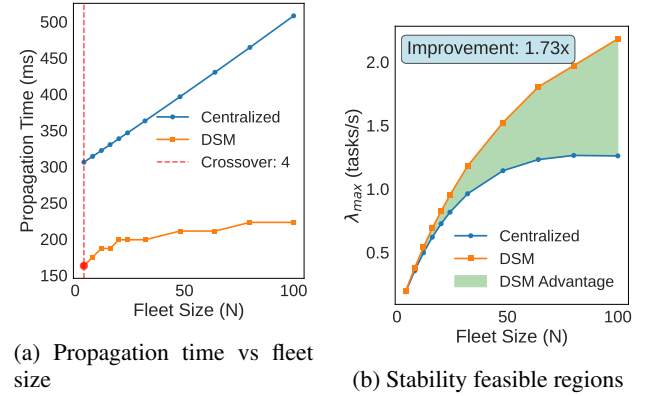


Figure 2: Analytical comparison of Distributed vs Centralized Control

REFERENCES

- Leaderless replication for distributed systems. Technical Report EECS-2020-121, University of California, Berkeley, 2020.
- Agaskar, A., Siva, S., Pickering, W., O’Brien, K., Kekeh, C., Li, A., Gallo Sarker, B., Chua, A., Nemade, M., Thattai, C., Di, J., Iyengar, I., Dharoor, R., Kirouani, D., Erskine, J., Hegazy, T., Niekum, S., Khan, U. A., Pecora, F., and Durham, J. W. Deepfleet: Multi-agent foundation models for mobile robots. *arXiv preprint arXiv:2508.08574*, 2025. doi: 10.48550/arXiv.2508.08574. URL <https://doi.org/10.48550/>

arXiv:2508.08574. Unpublished technical report, Amazon Robotics.

Beckers, R., Holland, O. E., and Deneubourg, J.-L. From local actions to global tasks: Stigmergy and collective robotics. *Artificial Life*, 4:181–189, 1994.

Dorigo, M. and Stützle, T. *Ant Colony Optimization*. MIT Press, Cambridge, MA, 2003.

Ilager, S., Fahringer, J., Tundo, A., and Brandić, I. A decentralized and self-adaptive approach for monitoring volatile edge environments. *arXiv preprint arXiv:2405.07806*, 2024. Submitted to ACM Transactions on Autonomous and Adaptive Systems.

Kingman, J. F. C. The single server queue in heavy traffic. *Mathematical Proceedings of the Cambridge Philosophical Society*, 57(4):902–904, 1961.

Mathur, A., Cavallaro, A., Padeleris, P., and Kate, B. Programming micro-aerial vehicle swarms with karma. In *Proceedings of the 9th ACM Conference on Embedded Networked Sensor Systems*, pp. 121–134. ACM, 2011.

Nguyen, A. A. Scalable, decentralized multi-agent reinforcement learning methods inspired by stigmergy and ant colonies. *arXiv preprint arXiv:2105.03546*, 2021.

Rappel, O., Amir, M., and Bruckstein, A. M. Stigmergy-based, dual-layer coverage of unknown indoor regions. In *Proceedings of the 2023 International Conference on Autonomous Agents and Multiagent Systems*, 2023. arXiv:2209.08573.

Sciascia, D., Pedone, F., and Gramoli, V. Leaderless consensus. *Distributed Computing*, 2021. ICDCS 2021.

Wang, Y. et al. Scalable atomic broadcast: A leaderless hierarchical algorithm. *Parallel Computing*, 118:103054, 2023.

Xu, X., Li, R., Zhao, Z., and Zhang, H. Stigmergic independent reinforcement learning for multi-agent collaboration. *arXiv preprint arXiv:1911.12504*, 2019.